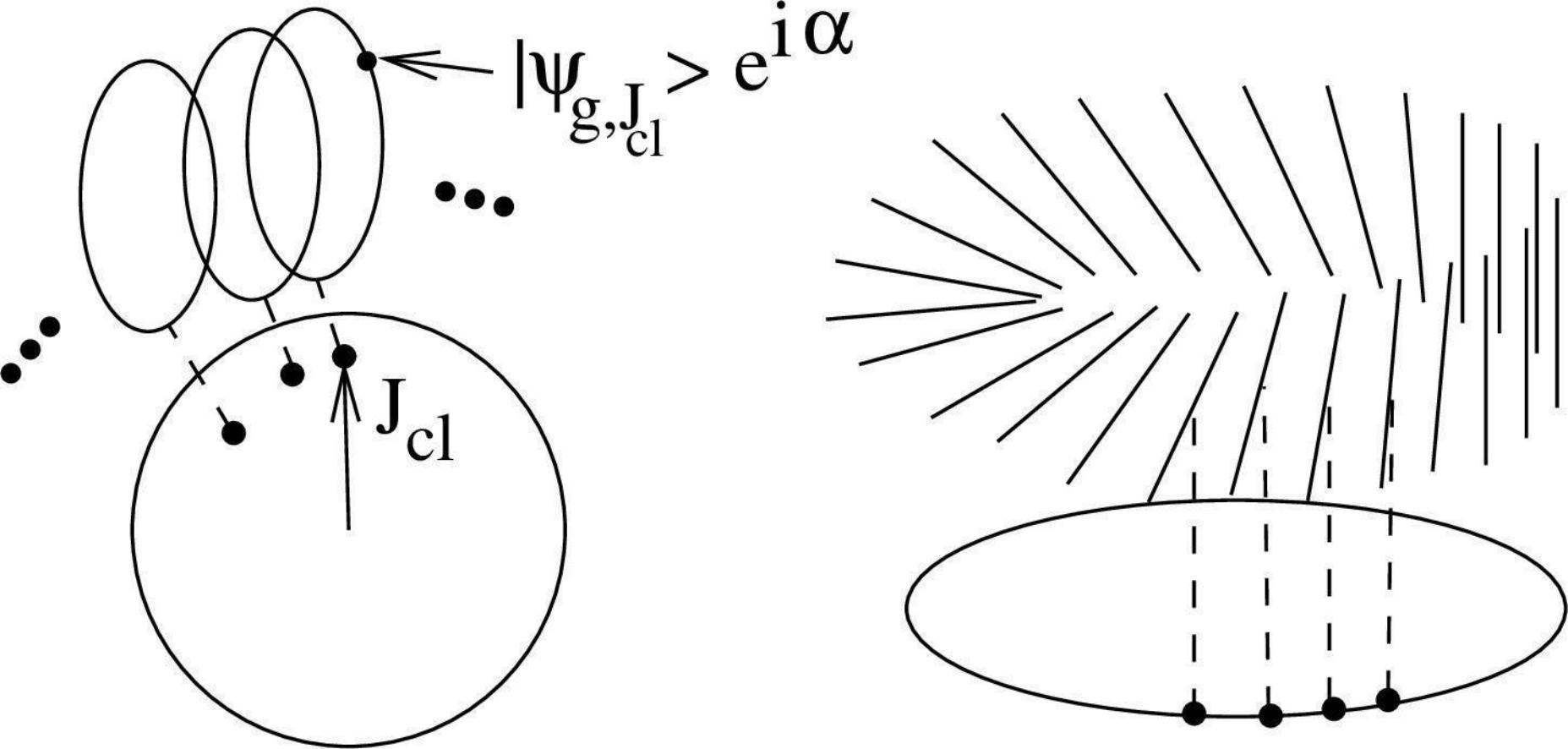

$|\psi_{g,J_{cl}}> e^{i\alpha}$
J_{cl}